

LOT (R) COMPUTER

HARDWARE RIG SPEC + BOM -- 3 July 2026

LOT SYSTEMS CORPORATION

LOT® COMPUTER — HARDWARE RIG SPECIFICATION

QUANTUM TILE / PAGER-CLASS AMBIENT NOTIFICATION DEVICE

DOCUMENT RIG-SPEC / LOT-COMPUTER-v1

ISSUE DATE 2026.07.03

CLASS INTERNAL / BUILD

STYLE TERMINAL GRID

VISION DOC docs/corporate/LOT-COMPUTER-VISION.md

INVENTOR Vadik Marmeladov · COSMO® CIA

00 FORM FACTOR — THE CONSTRAINT THAT DRIVES EVERY CHOICE

40mm x 40mm x 5mm · flat silver square · two-part stainless steel shell

Every component below is selected first for whether it can physically live inside a 5mm stack, second for cost at 100 units, third for being a proven, sourceable, off-the-shelf part (no custom silicon in v1). Where a part cannot fit flush, that is flagged here as a RISK, not hidden.

FACE A (polished) FACE B (function)

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Mirror-finish 316L E-paper display window

stainless steel plate. Camera lens boss

No seams, no branding, Piezo-disc button (under steel, no hole)

no ports. Qi coil alignment mark

01 COMPUTE + RADIO

ESP32-S3-WROOM-1 (N8R2 variant) — MCU + Wi-Fi + BLE

Dual-core Xtensa LX7, AI vector instructions, native 8-bit parallel camera interface (DVP), Wi-Fi 4 + BLE 5.0, deep-sleep <20uA. The camera interface being native to the SoC removes a second bridge chip from the 5mm stack — the single reason this part beats an NRF52 + separate Wi-Fi combo here.

DIGIKEY <https://www.digikey.com/en/products/detail/espressif-systems/ESP32-S3-WROOM-1-N8R2/15200058>

DATASHEET <https://www.espressif.com/en/products/socs/esp32-s3>

UNIT COST ≈ \$2.80-\$3.40 (qty 100)

02 DISPLAY — E-PAPER, NOT OLED

Good Display GDEY0154D67 — 1.54" 200x200, partial refresh, SSD1681

E-paper holds an image with **zero standing power draw** — the message

stays legible on the charging puck, unpowered, indefinitely. Partial-refresh mode (~0.3s) updates a single line without the full-frame black/white flash, which matters for the Ambient AI™ rule: *no alarm, one line, exact moment.*

Panel is ~1.2mm thick glass-on-glass — fits the 5mm stack with room for

the driver board underneath.

PRODUCT <https://www.good-display.com/product/388.html>

ALT MOD <https://www.buy-lcd.com/products/154-inch-electronic-paper-display-200x200-partial-refresh-digital-price-tags-screen-gdey0154d67>

UNIT COST ≈ \$6-\$9 (qty 100, direct factory quote required – no listed retail price)

RISK: Good Display prices via quote-only for panel-level orders. Get a formal 100-unit quote before locking BOM cost; do not assume retail-card pricing from hobbyist resellers.

03 CAMERA – THE HEIGHT-BUDGET RISK

OV2640, 2MP, DVP/FPC

ARDUCAM https://www.arducam.com/arducam-ov2640-camera-module-2mp-mini-ccm-compact-camera-modules-compatible-with-arduino_m0031esp32-esp8266-development-board-with-dvp-24-pin-interface_.html

FPC MODULE <https://www.ktron.in/product/ov2640-camera-module-with-24-pin-ffc-fpc/>

UNIT COST ≈ \$6-\$10 (qty 100)

RISK – FLAGGED, NOT HIDDEN:

Off-the-shelf OV2640 camera+lens assemblies run ~14mm x 14mm x ~5-8mm at the lens barrel alone. This does NOT fit flush inside a 5mm shell.

Two real options, both go into the roadmap as an open decision:

A. LENS BOSS – accept a small raised optical boss (~3-4mm) on Face B, the same way a phone accepts a camera bump. Body stays 5mm everywhere else. Matches how COSMO® hardware precedent already treats bioelectric sensor bosses. RECOMMENDED for v1.

B. BARE DIE + FPC – source a bare CoB (chip-on-board) sensor die on a flex cable with the lens assembly relocated to sit inside the charging puck instead of the tile (the puck has no 5mm constraint). Tile becomes presence-only (ambient light + IR proximity); full camera lives in the dock. Cleaner industrial design, more integration risk, deferred to v2 evaluation.

v1 BUILD PLAN uses OPTION A. This is the single biggest mechanical risk in the BOM and is carried forward into the roadmap risk register.

04 SENSOR – WEATHER, OFF THE SHELF

Bosch BME680 – temperature / humidity / pressure / gas (VOC)

LGA package, 2.5mm x 2.5mm x 0.93mm – the one part in this BOM that disappears into the stack with zero height risk. Feeds the existing LOT® Weather widget directly; feeds the future Air Quality widget (see [docs/corporate/LOT-AMBIENT-AI-VISION.md](#)) via VOC/gas channel. PM2.5/CO2 laser particulate sensing is NOT attempted in the tile – those sensors need a fan and airflow channel that cannot fit 5mm. PM2.5 stays a dock/LOT® Station-class product, not this device. That split is intentional, not a shortfall.

DIGIKEY <https://www.digikey.com/en/products/detail/bosch-sensortec/BME680/7401317>

OEM PAGE <https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors-bme680/>

ADAFRUIT <https://www.adafruit.com/product/3660> (breakout, for bring-up/dev only)

UNIT COST ≈ \$4-\$6 (qty 100)

05 BUTTON – PIEZO DISC, NOT A MECHANICAL SWITCH

Copy button, tap-sense via piezo element under stainless steel

A 2-part stainless shell has no clean way to run a mechanical tact-switch plunger without a visible seam or hole. Instead: a piezo disc bonded to the INSIDE face of Face B stainless steel detects a tap through the metal itself – the button has no visible hole, no seam, no moving part. This also lets the same disc double as a haptic-confirmation buzzer (one soft tick on tap-received), which is the piezoelectric / haptic-feedback language already established for the Quantum Cube in the CQGS white paper ([docs/corporate/CQGS-WHITE-PAPER-SNAPSHOT.md](#)) – this is that concept, built.

DIGIKEY CATEGORY <https://www.digikey.com/en/products/filter/buzzer-elements-piezo-benders/160>

PUI AUDIO BENDERS <https://www.digikey.com/en/product-highlight/p/pui-audio/piezo-benders>

ADAFRUIT (dev) <https://www.adafruit.com/product/1740>

UNIT COST ≈ \$1-\$2 (qty 100)

06 WIRELESS CHARGING

TI BQ51013B – Qi (WPC v1.1/1.2) receiver IC + coil

Sits in the tile; a matching transmitter coil sits in the stainless charging puck (separate accessory, see §08). 20-QFN package, fits the stack easily.

TI PRODUCT PAGE <https://www.ti.com/product/BQ51013B>

DATASHEET <https://www.ti.com/lit/ds/symlink/bq51013b.pdf>

DIGIKEY <https://www.digikey.com/en/products/detail/texas-instruments/BQ51013BRHLT/3877679>

UNIT COST ≈ \$3.50-\$3.60 (qty 100, cut-tape)

07 BATTERY – BRIDGE CELL, NOT A DAILY-USE PACK

3.7V 150mAh LiPo (JST-PH)

ADAFRUIT <https://www.adafruit.com/product/1317>

UNIT COST ≈ \$5.95 (qty 1 retail; expect ~\$3-4 at qty 100 direct from cell OEM)

DESIGN NOTE: a 150mAh pouch cell alone runs ~4-5mm thick – close to the ENTIRE height budget by itself once you stack it against the PCB, e-paper panel, and camera boss. The resolution: the LOT® Computer is an ambient, stationary object that lives on its charging puck (§08) the overwhelming majority of the time, the same way the LOT® Station and LOT® Brush are dock-resident hardware, not portables. The on-board cell is a BRIDGE CELL sized for the seconds-to-minutes a person lifts or repositions the tile – not for all-day untethered operation. This directly resolves the 5mm constraint instead of pretending it away. Evaluate a smaller 40-80mAh cell in prototype fit-check before locking the BOM.

08 CHARGING PUCK (ACCESSORY, NOT HEIGHT-CONSTRAINED)

Matching stainless steel puck, Qi transmitter reference design built around a standard Qi Tx IC (e.g. TI BQ500212A family – spec at prototype stage, not locked here) plus mains adapter. The puck has no 5mm limit and is the natural home for anything that doesn't fit the tile: larger antenna, status LED (hidden, under-steel diffusion, same "no visible light" rule as the Ambient AI™ doctrine), and – per §03 Option B – a future camera relocation target if the tile-mounted lens boss proves undesirable in field testing.

09 ENCLOSURE – TWO-PART STAINLESS STEEL, PCBWAY

Both shells (Face A polished, Face B brushed with display/lens/coil cutouts) machined from 316L stainless (marine-grade, corrosion-resistant, the same grade already documented for CNC service at the fab partner). PCBWay is the single vendor for enclosure + PCB + SMT assembly on this run, keeping supply chain and QA under one accountable roof for a 100-unit pilot.

CNC STAINLESS (316L)

<https://www.pcbway.com/rapid-prototyping/cnc-machining/metal/stainless-steel/Stainless-steel-316-316L/>

CNC STAINLESS (304)

<https://www.pcbway.com/rapid-prototyping/cnc-machining/metal/stainless-steel/Stainless-steel-304/>

SMT PCB ASSEMBLY <https://www.pcbway.com/pcb-assembly.html>

ASSEMBLY CAPABILITIES https://www.pcbway.com/pcb_prototype/SMT_Assembly_Capabilities.html

QUOTE REQUEST <https://www.pcbway.com/rapid-prototyping/manufacture/?type=2>

UNIT COST (2-shell CNC 316L, polish + bead-blast finish) ≈ \$18-\$28 (qty 100)

UNIT COST (4-layer PCB, SMT assembly, turnkey) ≈ \$12-\$18 (qty 100)

10 BILL OF MATERIALS – 100-UNIT ROLLUP

COMPONENT	UNIT COST	x100	NOTE
ESP32-S3-WROOM-1-N8R2	\$3.20	\$320	MCU+radio, camera-native
GDEY0154D67 e-paper	\$7.50	\$750	quote-only, verify before lock
OV2640 camera + lens	\$8.00	\$800	lens-boss option A
BME680 weather/gas sensor	\$5.00	\$500	LGA, no height risk
Piezo disc (Copy button)	\$1.50	\$150	doubles as haptic buzzer
BQ51013B Qi receiver	\$3.55	\$355	+ coil + support caps
150mAh LiPo bridge cell	\$3.50	\$350	verify smaller cell in fit-check
PCB (4L, bare)	\$4.00	\$400	via PCBWay
SMT assembly (per unit)	\$12.00	\$1,200	via PCBWay turnkey
CNC stainless shell pair	\$22.00	\$2,200	316L, polish + brushed finish
Passives / misc SMD	\$2.50	\$250	caps, resistors, ESD protection
Wireless charging puck	\$9.00	\$900	Qi Tx + mains adapter, accessory
Packaging + manual insert	\$3.00	\$300	box, foam, printed quick-start
SUBTOTAL / UNIT	≈ \$85.25		
SUBTOTAL / 100-UNIT RUN		≈ \$8,525	
NRE (tooling, CNC fixtures, first-article) – separate, ≈ \$2,000-\$4,000 one-time			

Prices are current street/distributor estimates pulled July 2026 – verify against a live PCBWay quote and distributor cart before committing purchase orders. The e-paper panel in particular is quote-gated, not retail-priced.

11 WHAT THIS SPEC DELIBERATELY DOES NOT DO

- No custom ASIC, no custom sensor silicon – every part above is sourceable off a distributor shelf today ("AI-grade off-the-shelf

sensors," per the intake brief).

- No injection-molded plastic tooling – CNC stainless only, appropriate for a 100-unit pilot; injection molding is a v2+ decision once volume justifies tooling amortization.
- No PM2.5/particulate sensing in the tile – deliberately deferred to a dock-class product (§04, §08).

LOT SYSTEMS CORPORATION LOS ANGELES, CA

FIRMWARE SPEC → docs/technical/LOT-COMPUTER-FIRMWARE.md

SOFTWARE SPEC → docs/technical/LOT-COMPUTER-SOFTWARE-BRIDGE.md

ROADMAP → docs/technical/LOT-COMPUTER-ROADMAP.md

END OF SPECIFICATION 2026.07.03